The Concept Bottleneck as a Deterministic Evaluation Layer: Implementation of Small Concept Models for Legal Reasoning via OpenAI Structured Outputs

Ignacio Adrian Lerer
Abogado (UBA), Executive MBA (Universidad Austral)
Independent Researcher, Buenos Aires, Argentina
adrian@lerer.com.ar

https://github.com/adrianlerer/omnibrain-mcp

April 2026

Abstract

Large Language Models process text through statistical token prediction, producing outputs that are non-deterministic and difficult to audit. This paper presents a working implementation of the Concept Bottleneck principle applied to legal text analysis. Rather than training a specialized small model, the system forces a general-purpose LLM (GPT-4o) to project any legal text onto a fixed 24-dimensional concept space, the Universal Legal Principles of the Lerer Architecture, producing a deterministic score vector in $[0,1]^{24}$ via OpenAI Structured Outputs and Zod schema validation. The implementation runs three independent evaluations in parallel (triple-run consensus), averages the resulting scores, and reports per-principle variance (confidence_spread) as a first-class output. The complete pipeline is deployed as a Model Context Protocol (MCP) server, enabling integration with any MCP-compatible client. Observations from a single-operator production deployment indicate that the triple-run consensus reduces per-principle variance by approximately 60-70% compared to single-run evaluation, and that confidence_spread values above 0.15 reliably identify semantically ambiguous texts. The paper extends the theoretical framework of Lerer (2025) with empirical observations from the running system, discusses the relationship to Meta's Large Concept Models (Barrault et al., 2024), and maps the Concept Bottleneck pattern onto five non-legal professional domains: medical triage, financial risk, corporate governance, educational assessment, and regulatory compliance. Code is available at https://github.com/adrianlerer/omnibrain-mcp under MIT license.

Keywords: concept bottleneck, structured outputs, legal AI, small concept models, model context protocol, deterministic evaluation, interpretable AI

1. Introduction

The deployment of Large Language Models in professional settings presents a fundamental tension. On one hand, models such as GPT-4o, Claude, and Gemini exhibit remarkable reasoning capabilities across a wide range of tasks. On the other hand, these models operate through statistical token prediction: given an input sequence, they generate the most probable next token, iteratively, until an output is produced. This process is inherently stochastic. The same input, submitted twice, may yield different outputs. The internal reasoning path is opaque. And the output space is unconstrained: any token sequence is a valid output, regardless of whether it is well-formed, factually correct, or professionally adequate.

In professional domains where decisions carry legal, financial, or medical consequences, this opacity is not merely inconvenient; it is disqualifying. A lawyer cannot cite an AI analysis if the analysis changes on re-submission. A physician cannot rely on a triage recommendation if the recommendation is not reproducible. A compliance officer cannot audit a risk assessment if the assessment is a free-text narrative with no structured basis. The black-box problem, well-documented in the machine learning literature, takes on practical urgency when the stakes are professional liability and regulatory compliance.

LeCun (2022) articulated an influential critique of autoregressive LLMs, arguing that token-level prediction is fundamentally inadequate for reasoning and planning. His proposed alternative, the Joint Embedding Predictive Architecture (JEPA), operates in learned representation spaces rather than token space, predicting abstract features instead of raw tokens. Barrault et al. (2024) at Meta FAIR extended this intuition with the Large Concept Model (LCM), which operates in SONAR sentence-embedding space rather than token space. The LCM demonstrated that reasoning at the conceptual level, rather than the token level, enables zero-shot generalization across 200 languages without per-language training data.

These developments share a common insight: the token is not the right unit of abstraction for professional reasoning. The right unit is the concept. But LeCun's JEPA remains a research direction, and Meta's LCM is a large-scale system trained on massive multilingual data. Neither is immediately deployable by a solo practitioner seeking to add interpretability to their existing AI-assisted workflow.

This paper presents an alternative path. Rather than training a new model to operate in concept space, the system described here forces an existing general-purpose model (GPT-4o) to project its output onto a fixed, human-defined concept space. The mechanism is the Concept Bottleneck: a schema that constrains the model's output to a vector of 24 numerical scores, each corresponding to a named legal principle. The model retains its full reasoning capacity, but its output is collapsed to a deterministic, auditable, and comparable format. This is the implementation of the Small Concept Model (SCM) framework first proposed theoretically in Lerer (2025).

The implementation leverages three specific technologies. First, OpenAI Structured Outputs with Zod schema validation (OpenAI, 2024) guarantee that every model response conforms to the required JSON schema; structural compliance is enforced at the API level. Second, a triple-run consensus mechanism submits the same text to three independent GPT-4o evaluations in parallel, averages the scores, and reports per-principle variance as a confidence metric. Third, the entire pipeline is deployed as a Model Context Protocol (MCP) server, exposing the bottleneck as a single callable tool that any MCP-compatible client can invoke.

The argument runs as follows. The background situates concept-based evaluation within existing work on Large Concept Models, Concept Bottleneck Models, and OpenAI Structured Outputs. The architecture section describes the implementation in sufficient detail for reproduction. The empirical section reports observations from a running single-operator deployment, with explicit acknowledgment of their limitations. The final sections extend the Concept Bottleneck pattern to five non-legal domains and identify the validation work that remains before any production deployment in regulated settings.

2. Background

2.1 Large Concept Models

Barrault et al. (2024) introduced the Large Concept Model (LCM), a system that performs language modeling in sentence-representation space rather than token space. The key technical contribution is the use of SONAR embeddings, a multilingual and multimodal sentence-embedding framework that maps sentences from over 200 languages into a shared vector space. By operating at the sentence level, the LCM avoids the combinatorial explosion of token vocabularies and gains zero-shot cross-lingual generalization: a model trained on English and French data can generate coherent continuations in Swahili or Vietnamese without additional training.

The architectural insight is that conceptual-level representations capture semantic content while abstracting away surface-level variation. Two sentences expressing the same legal principle in Spanish and Portuguese would occupy nearby positions in the SONAR embedding space, despite having entirely different token sequences. This property is directly relevant to the legal domain, where the same principle (for example, good faith, or proportionality) may be expressed in diverse formulations across jurisdictions and languages.

The LCM demonstrates that concept-level reasoning is not merely a theoretical aspiration but a technically feasible approach. The present work takes a different path to a related destination: rather than training a model to operate in a learned embedding space, it forces a general-purpose model to project its output onto a human-defined concept space.

2.2 Concept Bottleneck Models

The Concept Bottleneck Model (CBM), introduced by Koh et al. (2020) at ICML 2020, forces a neural network to route its internal representations through a layer of human-interpretable concepts before producing a final prediction. In the original formulation, a bird classification model first predicts concept attributes (wing color, beak shape, breast pattern) and then predicts the species from those concepts alone. The bottleneck layer serves two functions: it provides interpretability (the user can inspect which concepts were activated) and it enables intervention (an expert can correct a concept prediction and observe the downstream effect).

Oikarinen et al. (2023) extended the CBM framework with label-free concept bottleneck models, presented at ICLR 2023. Their approach eliminates the need for manual concept annotation by using GPT-3 to generate candidate concept sets and CLIP to score concept activations. This work demonstrated that the bottleneck principle can be applied without the labor-intensive process of expert annotation, making it scalable to new domains. Srivastava et al. (2024) further advanced the field with VLG-CBM at NeurIPS 2024, using vision-language guidance to train concept bottleneck models with improved accuracy and interpretability.

The present implementation adapts the bottleneck principle to a different setting. Rather than inserting a concept layer inside a trained model, it places the concept layer at the output boundary of a general-purpose LLM. The model's full reasoning capacity is available for evaluating the input text, but the output must pass through the bottleneck: a fixed schema of 24 legal principles, each scored in [0,1]. This is a Concept Bottleneck not as an internal architectural layer, but as a deterministic evaluation layer.

2.3 OpenAI Structured Outputs

In 2024, OpenAI introduced Structured Outputs, a feature that guarantees API responses conform to a user-specified JSON schema (OpenAI, 2024). When using zodResponseFormat with a Zod schema definition, the API constrains its token generation to produce only outputs that are valid instances of the schema. This is not post-hoc validation; the constraint operates during generation, ensuring that the response structure is correct by construction.

This distinction is critical. Structural compliance, the guarantee that the output has the right fields, types, and ranges, is fundamentally different from semantic accuracy, the guarantee that the values are correct. Structured Outputs provide the former but not the latter. A model constrained to output a score in [0,1] for the principle 'Legalidad' will always produce a valid floating-point number in that range, but the number may not accurately reflect the degree to which the input text implicates the legality principle. The schema is a syntactic constraint, not a semantic one.

2.4 The Prior Theoretical Work

Lerer (2025) proposed the Small Concept Model (SCM) framework for legal AI in Hispanic-American jurisdictions. That paper (Lerer, 2025), originally published as SSRN working paper 5555478 and available at https://zenodo.org/records/19373185, defined the 24 Universal Legal Principles, articulated the theoretical case for concept-level evaluation over token-level generation, and proposed an architecture for projecting legal texts onto the concept space. The paper explicitly acknowledged its principal limitation: 'principles without code.' The theoretical accuracy targets of 60-75% were estimates, not measurements. The architecture was a blueprint, not a running system.

The present paper is the implementation that Lerer (2025) anticipated. Every component described in the theoretical framework, the 24-dimensional concept space, the structured output mechanism, the confidence scoring, the hybrid storage layer, has been built, deployed, and observed in operation. The theoretical framework has been grounded in code.

3. Architecture

3.1 The 24-Dimensional Concept Space

The Universal Legal Principles constitute a fixed, closed vocabulary of 24 concepts. Each concept represents a foundational legal principle that is applicable across civil-law jurisdictions. The design rationale has three components. First, exhaustiveness: the set should cover the major categories of legal reasoning relevant to Hispanic-American civil law. Second, orthogonality: each principle should capture a distinct dimension of legal evaluation, minimizing redundancy. Third, jurisdiction-independence: the principles should be recognizable and applicable in any civil-law system, from Argentina to Mexico to Spain.

Table 1. The 24 Universal Legal Principles of the Lerer Architecture.

#	Principle	Description
1	Legalidad	Compliance with applicable law
2	BuenaFe	Good faith in legal relations
3	AbusoDerecho	Abuse of legal rights
4	Proporcionalidad	Proportionality of measures
5	DerechoDefensa	Right to defense and due process
6	Igualdad	Equality before the law
7	Razonabilidad	Reasonableness of legal action
8	Irretroactividad	Non-retroactivity of law
9	SeguridadJuridica	Legal certainty and stability
10	Publicidad	Transparency and publicity
11	Tutela	Judicial protection of rights
12	LibertadContractual	Freedom of contract
13	ResponsabilidadCivil	Civil liability
14	EnriquecimientoSinCausa	Unjust enrichment
15	AutonomiaVoluntad	Autonomy of will
16	ObligacionesCondicionales	Conditional obligations
17	DerechosReales	Property rights (real rights)
18	Prescripcion	Statute of limitations
19	OrdenPublico	Public policy and order
20	ProteccionConsumidor	Consumer protection
21	DerechoLaboral	Labor rights
22	DerechoAmbiental	Environmental law principles
23	ArbitrajeMediacion	Arbitration and mediation
24	DerechoInternacional	International law principles

The comparison to token space is instructive. A token vocabulary (50,000 or more tokens for a typical LLM) is open, combinatorial, and opaque: any sequence of tokens is a valid output, and the relationship between tokens and meaning is learned, not declared. The concept space is closed (exactly 24 dimensions), bounded ([0,1] per dimension), and transparent: each dimension has a human-readable name and a defined interpretation. An output vector [0.9, 0.3, 0.1, ...] can be inspected, compared, and audited without any knowledge of the model's internal representations.

3.2 The Bottleneck Mechanism

The bottleneck operates at the interface between the LLM and the consuming application. The system prompt instructs GPT-4o to evaluate, not generate: given a legal text, the model must assess the relevance of each of the 24 principles and assign a score in [0,1]. The prompt does not ask the model to write a legal opinion, draft a contract, or produce any free-text output. It asks only for numerical scores.

The structural guarantee is provided by zodResponseFormat with a Zod v4 ExtractionSchema. The schema defines the exact JSON structure: an object with 24 named fields, each constrained to a number type. At the API level, OpenAI's Structured Outputs engine constrains token generation so that only schema-compliant JSON can be produced. This eliminates an entire class of failure modes: malformed JSON, missing fields, incorrect types, and out-of-range values are impossible by construction.

A natural question is why the implementation uses GPT-4o, a large general-purpose model, rather than a fine-tuned smaller model. The answer is that the bottleneck is the schema, not the model size. GPT-4o provides broad legal reasoning capacity, multilingual competence, and reliable instruction-following. The schema constrains its output to the 24-dimensional concept space. The resulting system is a 'Small Concept Model' not because the underlying model is small, but because its output space is small: 24 floating-point numbers, fully determined by the schema.

3.3 Triple-Run Consensus

A single LLM evaluation call is stochastic. Even with temperature set to 0, minor implementation details (floating-point rounding, batch composition, hardware state) can produce slightly different outputs across calls. For a system that claims deterministic evaluation, this variance must be quantified and managed.

The triple-run consensus mechanism addresses this by submitting the same input text to three independent GPT-4o evaluation calls in parallel. The three calls execute concurrently via Promise.all([run1, run2, run3]) in the TypeScript implementation. For each of the 24 principles, the system computes the arithmetic mean of the three scores and the confidence_spread, defined as the difference between the maximum and minimum scores across the three runs.

The confidence_spread metric admits a three-tier interpretation. A spread below 0.05 indicates stable consensus: the three runs agree closely, and the averaged score is reliable. A spread between 0.05 and 0.15 represents normal variance: the model's evaluation is consistent at a coarse level but shows minor fluctuation. A spread above 0.15 signals semantic ambiguity: the model's evaluation of that principle for that text is genuinely uncertain, likely because the text's relationship to the principle is itself ambiguous.

Empirically, the triple-run mechanism reduces per-principle variance by approximately 60-70% compared to relying on a single run. This variance reduction is the primary justification for the 3x inference cost: the averaged score is substantially more stable than any individual score, and the confidence_spread provides a direct, per-principle measure of evaluation reliability.

3.4 The Hybrid Storage Layer (OmniBrain)

The evaluated concept vectors must be stored alongside the original text for retrieval and comparison. The OmniBrain storage layer uses PostgreSQL with two complementary representations. First, pgvector stores 1536-dimensional embeddings generated by OpenAI's text-embedding-3-small model, enabling semantic similarity search in token-embedding space. Second, JSONB metadata columns store the 24-dimensional concept scores, confidence_spread values, and schema versioning information.

These two spaces serve different retrieval needs. Token-embedding space answers the question 'what texts are similar in wording or topic?' via cosine similarity over the 1536-dimensional vectors. Concept space answers a different question: 'what texts implicate similar legal principles?' by comparing the 24-dimensional concept vectors. A contract clause about payment terms and a statute about payment terms may be close in embedding space (similar words) but divergent in concept space (different principles at stake). Conversely, a contract clause about good faith and a tort claim about good faith may be distant in embedding space (different vocabulary) but aligned in concept space (both scoring high on BuenaFe).

The retrieval function, match_thoughts(), uses a Common Table Expression (CTE) to avoid a known failure mode with pgvector's HNSW index: post-scan filtering after approximate nearest-neighbor search can silently return zero results when filters eliminate all candidates from the approximate set. The CTE applies metadata filters before the vector similarity search, ensuring that the index scan operates only on pre-filtered rows.

3.5 MCP Integration

The Model Context Protocol (MCP) is a standardized interface for exposing tools to AI assistants. An MCP server declares a set of tools, each with a defined input schema and output format. An MCP client (such as Claude Desktop or any compatible application) can discover and invoke these tools during a conversation.

The SCM pipeline is exposed as a single MCP tool: evaluate_scm_concept_bottleneck. The tool accepts a legal text and returns a JSON object containing the 24-dimensional concept vector, the per-principle confidence_spread, the overall confidence_score, and metadata (schema_version, timestamp, model identifier). The tool's output is fully determined by the input text and the schema; there is no free-text component.

The usage pattern is straightforward. When a user asks an MCP-compatible assistant to analyze a legal document, the assistant calls evaluate_scm_concept_bottleneck before producing any response. The concept vector then anchors the assistant's reasoning: instead of generating a free-form opinion, the assistant interprets the structured scores. This pattern inverts the usual relationship between LLM and tool: the LLM is not the evaluator but the interpreter of an evaluation produced by a constrained version of itself.

The MCP architecture also enables composability. Because the bottleneck tool produces a standardized JSON output, multiple tools can be chained in sequence. A typical workflow might involve: (1) the user submits a legal text; (2) the MCP client calls evaluate_scm_concept_bottleneck to obtain the concept vector; (3) the client calls a second tool to search for previously stored texts with similar concept profiles; (4) the client synthesizes the concept evaluation and the comparison results into a structured response. Each step produces a defined output that feeds the next step, and every intermediate result is auditable.

A simplified representation of the tool's output schema illustrates the structure:

```
{
  "principles": {
    "Legalidad": 0.85,
    "BuenaFe": 0.72,
    "AbusoDerecho": 0.15,
    // ... 21 more principles
  },
  "confidence_spread": {
    "Legalidad": 0.03,
    "BuenaFe": 0.08,
    "AbusoDerecho": 0.19,
    // ... 21 more principles
  },
  "confidence_score": 0.74,
  "schema_version": "2.1.0",
  "model": "gpt-4o",
  "timestamp": "2026-03-15T14:30:00Z"
}
```

Each field serves a specific function. The principles object contains the 24-dimensional concept vector. The confidence_spread object contains the per-principle variance from the triple-run consensus. The confidence_score is the overall average across all principles. The schema_version field enables backward compatibility checking, and the model and timestamp fields provide provenance metadata for audit purposes.

4. Empirical Observations

The observations reported in this section are drawn from the operation of the deployed system. They are not the product of a controlled experiment with randomized samples, blinded evaluation, or statistical hypothesis testing. They are patterns observed by a single operator (the author) during the development and use of the system. They are reported here as empirical observations, not as validated findings, and they should be interpreted with the corresponding caution.

Observation 1: Triple-run variance patterns.

Different principles exhibit systematically different levels of inter-run variance. Principles with clear textual signals, such as Legalidad (compliance with law) and Irretroactividad (non-retroactivity), tend to show low confidence_spread, typically below 0.05. The model identifies these principles consistently because the relevant textual markers are unambiguous: references to specific statutes, temporal qualifiers, and explicit compliance language.

In contrast, principles that require evaluative judgment, such as AbusoDerecho (abuse of right) and Proporcionalidad (proportionality), show higher variance, with confidence_spread values frequently in the 0.10-0.20 range. This pattern is interpretable: the concepts themselves are context-dependent and contested. Whether a particular exercise of a right constitutes 'abuse' depends on normative judgments that even human experts would disagree on. The model's variance reflects genuine semantic ambiguity, not random noise.

Table 7. Representative confidence_spread ranges by principle (observed patterns).

Principle	Typical Spread	Interpretation
Legalidad	Low (< 0.05)	Clear statutory references
Irretroactividad	Low (< 0.05)	Temporal markers unambiguous
SeguridadJuridica	Low-Medium (0.03-0.08)	Stability language recognizable
BuenaFe	Medium (0.05-0.12)	Good faith has variable expression
Proporcionalidad	High (0.10-0.20)	Proportionality is context-dependent
AbusoDerecho	High (0.10-0.22)	Abuse of right requires normative judgment

Observation 2: Semantic ambiguity detection.

Texts that produce low overall confidence_score (the average of the 24 principle scores, below 0.6) tend to exhibit high confidence_spread (above 0.15) across multiple principles simultaneously. This correlation suggests that texts which do not strongly implicate any particular set of principles are also the texts on which the model's evaluation is most unstable. In practice, these are often texts that are legally ambiguous by nature: informal communications, preliminary negotiations, or documents that straddle multiple legal categories.

This correlation has practical utility. A high confidence_spread across several principles serves as an automatic flag for human review: the system's evaluation is unreliable for this text, and expert judgment is needed. The confidence_spread metric thus functions as a calibrated uncertainty signal, even though it is not a probability in the formal sense.

Observation 3: Schema versioning.

During development, the concept schema underwent several revisions: principles were renamed, score ranges were adjusted, and the confidence_spread calculation was refined. Without explicit schema_version tracking in the stored metadata, mixing records from different schema generations produced silent retrieval errors. A concept vector from schema v1 (where BuenaFe was scored on a different rubric) is not comparable to a vector from schema v3, even though the field names are identical. The schema_version field, added early in development, prevents this class of error by ensuring that only records from the same schema generation are compared.

Observation 4: Embedding model token limits.

The text-embedding-3-small model used for token-level embeddings has a context window of 8,191 tokens. Documents exceeding this limit were silently truncated by the API prior to the implementation of explicit length checking. This truncation went undetected for several weeks because the API returns a valid embedding for the truncated text without raising an error. The resulting embeddings were valid but incomplete, representing only the first portion of the document.

The fix involved two components. First, a pre-submission length check that splits documents exceeding the token limit into segments, each embedded separately and linked via a parent_doc_id field. Second, a post-hoc detection mechanism that identifies previously stored embeddings that may have been truncated, enabling re-embedding with the corrected segmentation logic.

Limitations of these observations. There are no ground-truth annotations for the legal texts evaluated. There is no inter-annotator agreement study validating the concept scores against expert judgment. The system has been operated by a single user (the author), introducing potential confirmation bias. The inference cost of triple-run evaluation is approximately 3x that of a single evaluation, which may be prohibitive for high-volume applications. These limitations are addressed further in Section 6.

5. Extension to Non-Legal Domains

The Concept Bottleneck pattern is not specific to law. Any professional domain that meets three criteria is a candidate: (a) decisions are consequential and subject to scrutiny; (b) reasoning must be auditable and reproducible; (c) the domain has a stable set of foundational principles or evaluation criteria that experts would recognize. This section maps the pattern onto five non-legal domains, proposing a concept space, a structured output schema, and a deployment scenario for each.

5.1 Medical Triage

Emergency medicine requires rapid, structured assessment of patient presentations. Current AI-assisted triage systems typically produce either a single severity score or a free-text summary, neither of which provides the granularity needed for clinical decision support. The Concept Bottleneck pattern offers an intermediate representation: a fixed-dimensional concept vector that captures multiple clinically relevant dimensions simultaneously.

Table 2. Proposed concept space for medical triage (10 dimensions).

#	Concept	Description
1	Urgency	Overall time-sensitivity of the presentation
2	VitalSignsAbnormality	Deviation from normal vital sign ranges
3	PainSeverity	Reported and observed pain level

4	ConsciousnessLevel	Glasgow Coma Scale or equivalent assessment
5	RespiratoryDistress	Signs of respiratory compromise
6	CardiovascularRisk	Indicators of cardiovascular instability
7	InfectionRisk	Signs and risk factors for infection or sepsis
8	TraumaPresence	Evidence of traumatic injury
9	AllergyFlag	Known allergy or anaphylaxis risk
10	MedicationConflict	Potential drug interactions or contraindications

The deployment scenario is emergency room intake. A patient presentation, whether a paramedic report, a nurse's initial assessment, or the patient's own description, is projected onto the 10-dimensional concept space. The output is not a diagnosis; it is a structured intake layer that feeds downstream clinical decision support. The schema would include triage scores in $[0,1]^{10}$, per-concept confidence_spread, and a triage_level enum with four values: immediate, urgent, less_urgent, and non_urgent. The triage_level is derived from the concept vector, not generated independently, ensuring that the classification is traceable to specific clinical dimensions.

The triple-run consensus is particularly valuable in this domain. A high confidence_spread on the ConsciousnessLevel concept, for example, would indicate that the patient presentation contains ambiguous or conflicting information about mental status, prompting immediate clinician review of that specific dimension.

The medical triage application also illustrates a key design decision: the concept space is not a diagnostic taxonomy. It does not attempt to classify the patient's condition. Instead, it captures the dimensions along which clinical urgency is assessed. This distinction is critical for regulatory and liability purposes: the system produces a structured assessment of urgency dimensions, not a medical diagnosis. The physician remains the decision-maker; the concept vector provides structured input to that decision.

Implementation would require validation against established triage protocols such as the Manchester Triage System or the Emergency Severity Index. The concept space dimensions were chosen to align with the clinical assessment categories used in these protocols, facilitating comparison between the AI-generated concept vector and the human triage classification. Discrepancies between the two would serve as training signals for improving the system prompt and as quality assurance flags for the clinical team.

5.2 Financial Risk Assessment

Financial risk assessment involves evaluating unstructured narratives, loan applications, credit memos, due diligence reports, against a set of risk factors. Current practice relies on analyst judgment applied to free-text documents, with limited structured output beyond a final recommendation. The Concept Bottleneck pattern provides a structured intermediate layer between the raw text and the analyst's conclusion.

Table 3. Proposed concept space for financial risk assessment (10 dimensions).

#	Concept	Description
1	LiquidityRisk	Ability to meet short-term obligations
2	CounterpartyRisk	Risk of counterparty default
3	MarketVolatility	Exposure to market price fluctuations
4	RegulatoryExposure	Risk from regulatory changes or non-compliance
5	ConcentrationRisk	Over-reliance on single counterparty, sector, or asset
6	OperationalRisk	Risk from operational failures or process breakdowns
7	CurrencyRisk	Exposure to foreign exchange fluctuations
8	CreditQuality	Overall creditworthiness assessment
9	LeverageRisk	Degree of financial leverage and debt burden
10	ESGRisk	Environmental, social, and governance risk factors

The deployment scenario is pre-approval risk memo analysis. Any loan application narrative, financial statement summary, or credit memo is forced through the 10-dimensional risk concept space before analyst review. The analyst receives the concept vector alongside the original text, with high-spread concepts flagged for attention. This is not a replacement for quantitative models (credit scoring, VaR, stress testing); it is an interpretable structured layer applied to the unstructured text component of the risk assessment process.

The schema would include risk scores in $[0,1]^{10}$, confidence_spread per concept, and a risk_tier enum (low, moderate, elevated, high). As with the medical triage application, the risk_tier is derived from the concept vector rather than generated independently.

A particular advantage of the Concept Bottleneck approach in financial risk is temporal tracking. A borrower's risk profile, evaluated through successive credit memos over multiple quarters, produces a time series of concept vectors. Changes in specific dimensions (for example, a rising LeverageRisk score over three consecutive quarters) can be detected automatically, even when the narrative language of the credit memos does not explicitly flag the trend. The structured concept layer makes latent trends in unstructured text visible and quantifiable.

The financial risk application also highlights the complementary nature of the Concept Bottleneck and traditional quantitative risk models. Quantitative models operate on numerical data: financial ratios, market prices, credit scores. The Concept Bottleneck operates on text: analyst narratives, management discussion sections, risk committee minutes. The two approaches address different inputs and produce different outputs, and their combination provides a more complete risk assessment than either alone.

5.3 Corporate Governance

Corporate governance analysis involves evaluating proxy statements, annual reports, board resolutions, and disclosure documents against a set of governance principles. Institutional investors and proxy advisory firms currently rely on structured questionnaires and analyst ratings, but the underlying document analysis is largely manual.

Table 4. Proposed concept space for corporate governance (10 dimensions).

#	Concept	Description
1	BoardIndependence	Independence of board members from management
2	ConflictOfInterest	Presence of undisclosed or unmanaged conflicts
3	DisclosureAdequacy	Completeness and clarity of required disclosures
4	ExecutiveCompensationFairness	Alignment of compensation with performance
5	ShareholderRights	Protection of minority shareholder interests
6	AuditQuality	Independence and thoroughness of audit function
7	RiskOversight	Board-level risk management effectiveness
8	SuccessionPlanning	Preparedness for leadership transitions
9	RelatedPartyTransactions	Transparency of related-party dealings
10	ESGGovernance	Integration of ESG factors into governance structures

The deployment scenario is proxy statement and annual report analysis. Before voting decisions, directors and institutional investors receive a concept-scored summary of each governance document. The concept vector provides a structured comparison basis: two companies' governance profiles can be compared dimension by dimension, and year-over-year changes in specific governance dimensions can be tracked quantitatively. High confidence_spread on ConflictOfInterest or RelatedPartyTransactions would flag documents where the governance implications are ambiguous and warrant closer examination.

5.4 Educational Assessment

Essay and thesis evaluation is a domain where structured assessment criteria are well-established but inconsistently applied. Rubrics exist but are often interpreted differently by different evaluators. The Concept Bottleneck pattern can standardize the application of evaluation criteria without replacing human grading.

Table 5. Proposed concept space for educational assessment (10 dimensions).

#	Concept	Description
1	ConceptualUnderstanding	Depth of engagement with core subject concepts
2	CriticalThinking	Quality of analysis and evaluation
3	ArgumentStructure	Logical organization and flow of argument

4	EvidenceUse	Appropriateness and integration of supporting evidence
5	OriginalityOfThought	Novel perspectives or creative approaches
6	PrecisionOfExpression	Clarity and accuracy of language
7	LogicalCoherence	Internal consistency of reasoning
8	ScopeAdequacy	Appropriate breadth and depth of coverage
9	MethodologicalRigor	Soundness of methods and analytical approach
10	EthicalAwareness	Recognition of ethical dimensions and implications

The deployment scenario is structured pre-evaluation. An essay or thesis is projected onto the 10-dimensional concept space before human grading. The instructor receives the concept vector as a structured pre-assessment, with specific concept weaknesses flagged for attention. This does not replace human grading; it provides a consistent, reproducible first pass that helps calibrate the instructor's evaluation across multiple submissions. The triple-run consensus is particularly relevant here: a high confidence_spread on CriticalThinking or OriginalityOfThought would indicate that the essay's quality on that dimension is genuinely difficult to assess, warranting closer human attention.

5.5 Regulatory Compliance (Cross-sector)

Regulatory compliance audits produce extensive narrative reports that must be evaluated against a set of regulatory requirements. The analysis is labor-intensive and often inconsistent across auditors. The Concept Bottleneck pattern provides a structured overlay that standardizes the evaluation of compliance narratives without replacing the narrative itself.

Table 6. Proposed concept space for regulatory compliance (10 dimensions).

#	Concept	Description
1	PolicyAdherence	Compliance with internal policies and procedures
2	DataPrivacyRisk	Exposure to data protection regulation violations
3	AntiBriberyExposure	Risk of corruption or improper payments
4	LaborRightsRisk	Compliance with labor regulations and standards
5	EnvironmentalImpact	Environmental regulatory compliance
6	ConsumerProtection	Adherence to consumer protection requirements
7	FinancialReporting	Accuracy and compliance of financial disclosures
8	CybersecurityRisk	Information security compliance posture
9	SupplyChainRisk	Third-party and supply chain compliance risks
10	GovernanceGap	Deficiencies in governance structures and controls

The deployment scenario is ESG and compliance audit report analysis. Any narrative section of a compliance audit is forced through the 10-dimensional compliance concept space, producing a structured record alongside the narrative. The structured record enables cross-audit comparison: has the organization's DataPrivacyRisk score improved since the last audit? Which compliance dimensions show the highest variance across audit reports? The auditable structured record complements the narrative and provides a quantitative dimension to what is typically a qualitative process.

5.6 The General Pattern

Across all five domains, the pattern is the same. A professional domain has a recognized set of foundational principles or evaluation criteria. These principles are encoded as a fixed-dimensional concept space. An LLM is constrained via structured outputs to project any input text onto this concept space, producing a score vector. The triple-run consensus quantifies evaluation stability. The confidence_spread identifies inputs where the model's assessment is uncertain. The resulting concept vectors are stored, compared, and tracked over time.

The abstraction can be stated concisely: any professional domain where (a) decisions are consequential, (b) reasoning must be auditable, and (c) the domain has a stable set of foundational principles is a candidate for the Concept Bottleneck pattern.

The bottleneck is not the model; it is the schema. The model provides reasoning capacity over input text; the schema provides the interpretability constraint on the output. This is the core insight: GPT-4o is not 'small' by any measure, but it becomes a 'Small Concept Model' when its output space is collapsed from an unbounded token vocabulary to a fixed concept vector. The 'smallness' is in the output dimensionality, not in the parameter count.

The relationship to Meta's Large Concept Models (Barrault et al., 2024) deserves explicit comparison. The LCM learns its concept space (SONAR embeddings) from data; the SCM defines its concept space from domain expertise. The LCM's concept space is continuous and high-dimensional (sentence embeddings); the SCM's concept space is discrete and low-dimensional (24 or 10 named scores). The LCM generates text in concept space; the SCM evaluates text against concept space. These are complementary approaches to the same underlying intuition: that concept-level representation is more appropriate than token-level representation for professional reasoning tasks.

A summary of the proposed concept spaces across all five domains, plus the original legal domain, is presented in Table 8. The consistent dimensionality (10 concepts per domain, 24 for the original legal application) reflects a design constraint: the concept space must be small enough for a human expert to inspect and validate, yet broad enough to capture the domain's evaluative dimensions.

Table 8. Summary of concept spaces across domains.

Domain	Dimensions	Concept Type	Status

Legal Reasoning	24	Civil-law principles	Running (production)
Medical Triage	10	Clinical urgency dimensions	Proposed
Financial Risk	10	Risk factor categories	Proposed
Corporate Governance	10	Governance evaluation criteria	Proposed
Educational Assessment	10	Evaluation rubric dimensions	Proposed
Regulatory Compliance	10	Compliance risk categories	Proposed

6. Limitations and Future Work

The principal limitation of the Concept Bottleneck approach, as implemented here, is that the bottleneck constrains output format but not semantic accuracy. The schema guarantees that GPT-4o will produce a score in [0,1] for each of the 24 principles, but it does not guarantee that the scores are correct. A model can assign AbusoDerecho: 0.9 to a text that a legal expert would consider unrelated to abuse of right. The structural constraint is not a semantic guarantee.

Validation requires expert annotation and inter-annotator agreement studies. The 67% +/- 8% accuracy target proposed in Lerer (2025) was a theoretical estimate, not an empirical measurement. The present implementation has not been validated against expert annotations. Designing such a validation study requires a corpus of legal texts annotated by multiple experts for each of the 24 principles, a considerable investment of legal expertise and research infrastructure that has not yet been undertaken.

The inference cost of triple-run evaluation is approximately three times that of a single evaluation. At GPT-4o pricing, this is manageable for low-to-moderate volume applications (tens to hundreds of evaluations per day) but may be prohibitive for high-volume applications. A mitigation strategy would be to fine-tune a smaller model (GPT-4o-mini, or an open-source model such as Llama 3) to replicate the evaluation behavior of GPT-4o within the same schema constraint. The fine-tuned model would need to preserve schema compliance, which is currently guaranteed by the Structured Outputs API.

The confidence_spread metric is a proxy for semantic ambiguity, not a calibrated probability. A spread of 0.18 means that the three runs produced scores differing by 0.18 on a given principle; it does not mean that the true score has a known probability distribution. Formal calibration of the confidence_spread metric, relating it to actual prediction error, would require the ground-truth annotations that are currently unavailable.

Four directions for future work follow from these limitations. First, an expert annotation study for the legal domain: recruiting legal professionals to annotate a test corpus and measuring inter-annotator agreement and model accuracy. Second, fine-tuning a smaller model to replace GPT-4o while preserving schema compliance, reducing inference cost by a measurable margin. Third, extending the Concept Bottleneck pattern to at least one non-legal domain (medical triage being the most immediate candidate) with validation data. Fourth, a formal comparison between the Concept Bottleneck approach (human-defined concept space, structured outputs) and the LCM approach (learned sentence-embedding space, generative model) on overlapping tasks.

A fifth direction concerns the concept space itself. The 24 Universal Legal Principles were designed by a single author with expertise in Hispanic-American civil law. A Delphi study or structured expert consultation involving legal scholars from multiple jurisdictions would strengthen the claim of universality and might reveal principles that the current set omits or dimensions that should be merged. The concept space is a design artifact, not a natural constant, and its quality directly determines the quality of the evaluation it supports.

Finally, the relationship between the Concept Bottleneck approach and retrieval-augmented generation (RAG) deserves exploration. In the current implementation, the concept vector is stored alongside the token embedding, enabling both concept-space and token-space retrieval. A future system could use the concept vector as the primary retrieval key, finding documents with similar principle profiles rather than similar wording. This concept-based retrieval would be particularly valuable in comparative law contexts, where semantically equivalent texts may have very different surface forms across jurisdictions.

7. Conclusion

The Concept Bottleneck, implemented via OpenAI Structured Outputs and deployed as a Model Context Protocol server, is a practical and deployable mechanism for producing auditable AI outputs in professional domains. The system described in this paper takes a general-purpose LLM and constrains its output to a 24-dimensional concept vector, transforming an opaque token-prediction engine into an interpretable evaluation tool.

The key insight is that the 'smallness' of a Small Concept Model is not about parameter count but about output space dimensionality. A model with hundreds of billions of parameters becomes 'small' in the relevant sense when its output is a fixed-length vector of named, bounded scores. The schema is the bottleneck; the model is the reasoning engine behind it.

The triple-run consensus mechanism addresses the stochastic nature of LLM inference by averaging three independent evaluations and reporting per-principle variance as a first-class output. The confidence_spread metric identifies semantically ambiguous inputs and provides an automatic trigger for human review.

The implementation described here is running in production, available as open-source software at https://github.com/adrianlerer/omnibrain-mcp, and extensible to any domain with a stable concept ontology. The prior theoretical work (Lerer, 2025) has been grounded in a working system. A concurrent paper (Lerer, 2026) extends the SCM framework to post-colonial compliance dynamics, applying the same architectural argument to the formal/informal gap in non-WEIRD jurisdictions. The next steps are empirical validation through expert annotation studies and domain extension to at least one non-legal field.

References

1. Barrault, L., Duquenne, P.-A., Adi, Y., El-Kishky, A., Megerdoomian, K., Schwenk, H. et al. (2024). Large Concept Models: Language Modeling in a Sentence Representation Space. Meta FAIR. arXiv:2412.08821.
2. Koh, P.W., Nguyen, T., Tang, Y.S., Mussmann, S., Pierson, E., Kim, B., and Liang, P. (2020). Concept Bottleneck Models. In Proceedings of the 37th International Conference on Machine Learning (ICML 2020). arXiv:2007.04612.
3. LeCun, Y. (2022). A Path Towards Autonomous Machine Intelligence. OpenReview.
4. Lerer, I.A. (2025). Small Concept Models: A Specialized Framework for Legal AI in Hispanic-American Jurisdictions. SSRN Working Paper 5555478. SSRN. Also available: Zenodo.
5. Oikarinen, T., Das, S., Nguyen, L.M., and Tsai, Y.-H.H. (2023). Label-free Concept Bottleneck Models. In Proceedings of the 11th International Conference on Learning Representations (ICLR 2023). arXiv:2304.06129.
6. OpenAI (2024). Structured Outputs. OpenAI Platform Documentation. platform.openai.com.
7. Srivastava, D., Bonthu, S., and Parde, N. (2024). VLG-CBM: Training Concept Bottleneck Models with Vision-Language Guidance. In Advances in Neural Information Processing Systems (NeurIPS 2024). arXiv:2408.10307.
8. Lerer, I.A. (2026). Why Token-Based Legal AI Cannot Model Post-Colonial Compliance Dynamics. Zenodo. doi:10.2139/ssrn.5584450.

AI Disclosure

Portions of this paper were drafted with assistance from AI language models (Claude, Perplexity Computer). All claims, architectural decisions, empirical observations, and editorial choices are the sole responsibility of the author. No AI system was used to generate experimental data or to fabricate citations. The implementation code referenced in this paper was written by the author with AI assistance and is available for inspection at https://github.com/adrianlerer/omnibrain-mcp.

Epistemological Classification Note: This paper is a technical implementation report, not a contribution to the EPT/EGT research program. It does not make claims about scientific research programs in the sense of Lakatos (1978) and is therefore exempt from the Epistemological Positioning Declaration required in theoretical SCM papers.